1. EXECUTIVE SUMMARY

Arogya Sathi is a comprehensive, AI-driven mobile ecosystem designed to address the specific "lifestyle disease" crisis in Urban India. Unlike generic fitness apps that offer standardized advice, Arogya Sathi functions as a **proactive "Virtual Nurse."** It operationalizes real-time environmental data (AQI, Heatwaves), bridges the linguistic divide using fine-tuned Indic AI models, and simplifies medical diagnostics via Computer Vision.

Crucially, the system is architected for **Data Sovereignty**, ensuring full compliance with the **Digital Personal Data Protection (DPDP) Act, 2023**, and is designed to integrate with the **Ayushman Bharat Digital Mission (ABDM)** ecosystem.

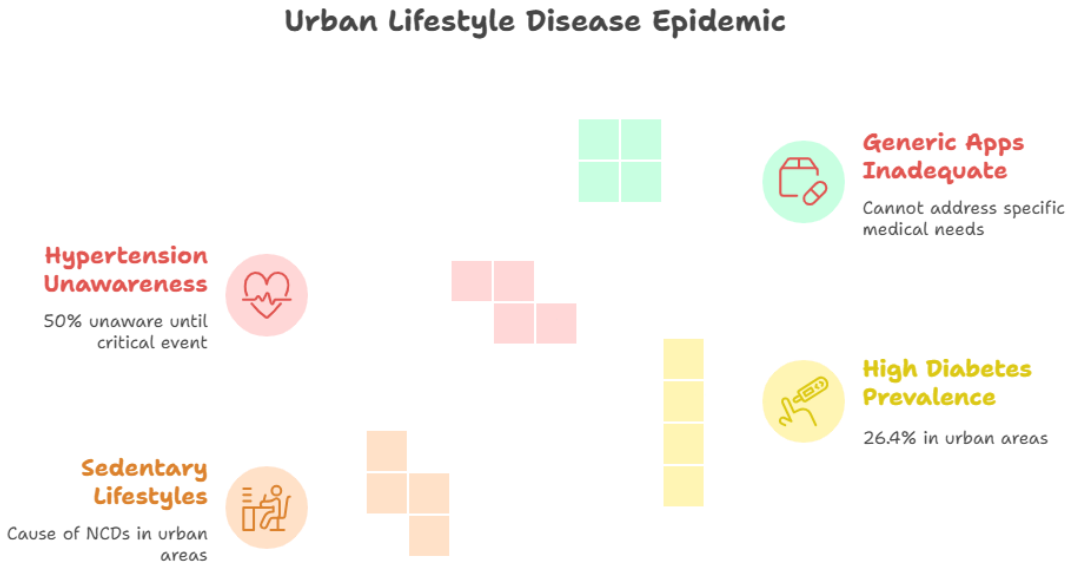
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2. PROBLEM STATEMENT (Evidence-Based)

We have identified four critical failures in the current healthcare technology landscape for Indian cities.

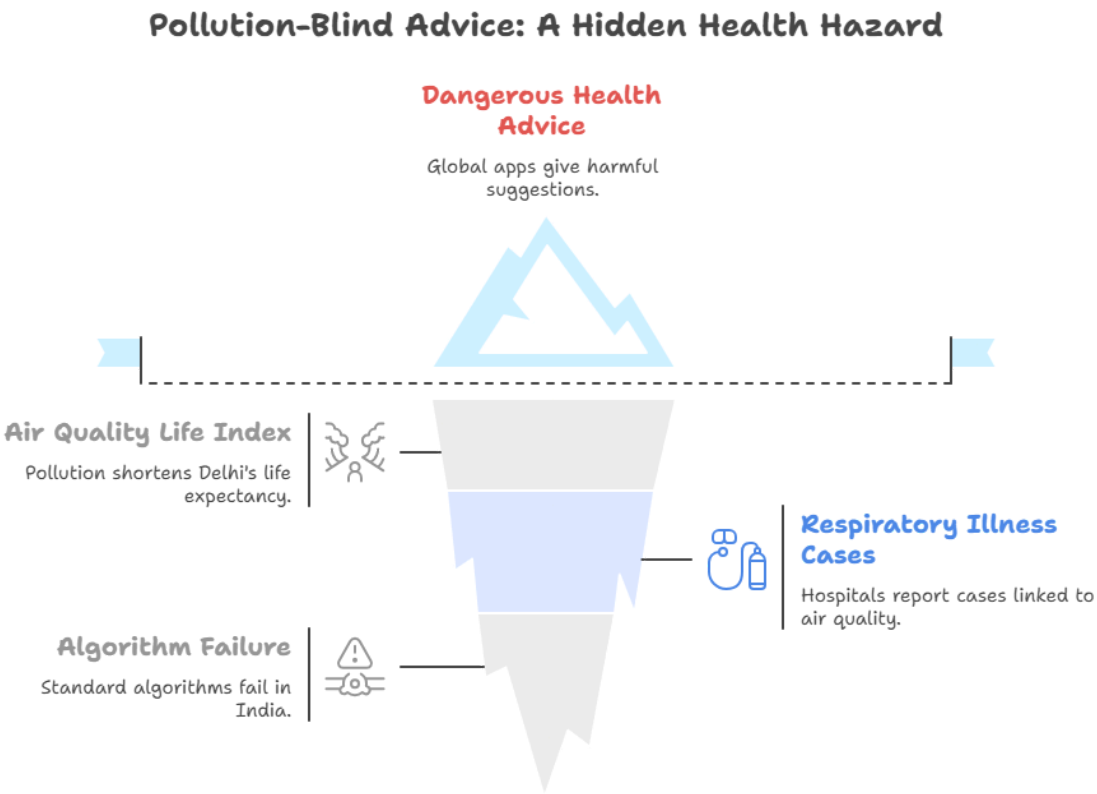
2.1. The Urban "Lifestyle Disease" Epidemic

- **The Problem:** Urban Indians are suffering from a "Silent Epidemic" of Non-Communicable Diseases (NCDs) due to sedentary lifestyles, yet digital tools are reactive, not preventative.
- **Research Evidence:**
  - According to the **ICMR-INDIAB Study (2023)**, the largest epidemiological study in India, the prevalence of diabetes in urban areas is **26.4%**, nearly double that of rural areas.
  - Hypertension affects nearly **1 in 4** urban adults, yet over 50% remain unaware of their condition until a critical event occurs.
- **Impact:** A generic step-counter app cannot address a 26% diabetes rate; specific, medical-grade lifestyle monitoring is required.



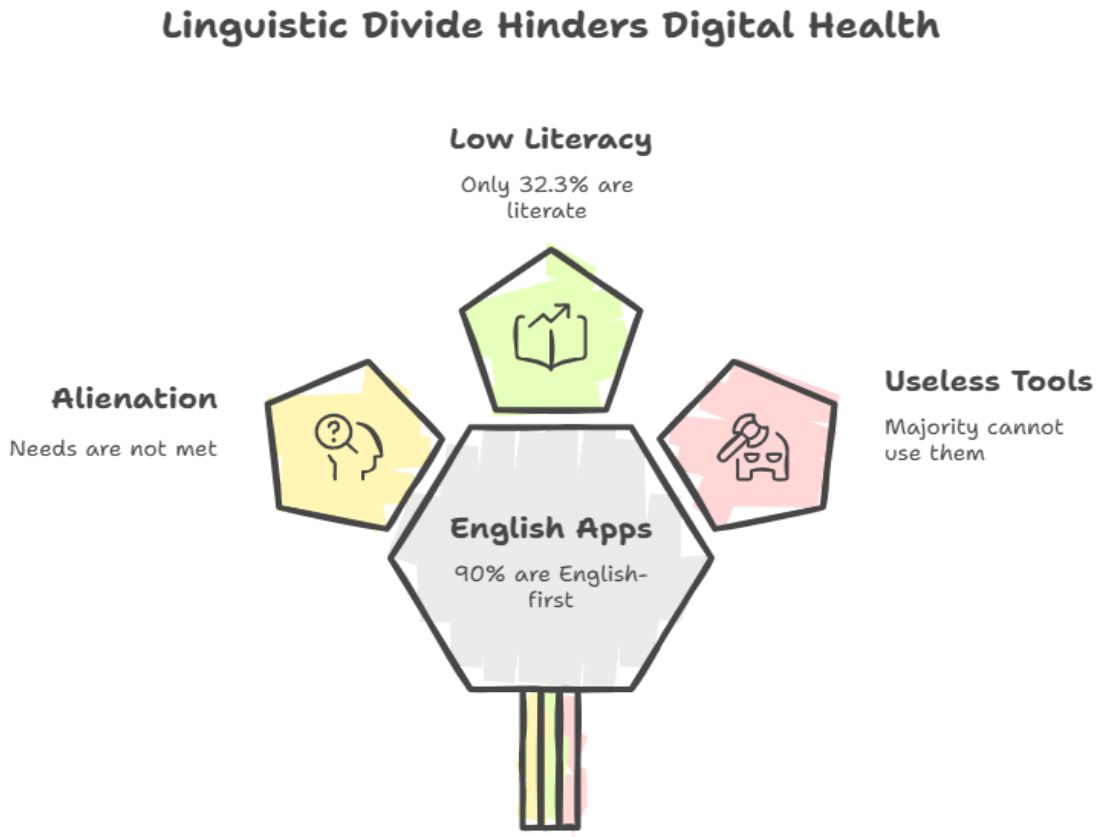
2.2. The "Pollution-Blind" Advice Gap

- **The Problem:** Global health apps often give dangerous advice in the Indian context (e.g., suggesting "morning runs" during toxic smog events).
- **Research Evidence:**
  - The **Air Quality Life Index (AQLI) 2024** reports that pollution in New Delhi shortens life expectancy by **11.9 years**.
  - In 2024, Delhi hospitals reported over **2 lakh emergency cases** of acute respiratory illness directly linked to air quality spikes.
- **Impact:** Standard algorithms fail in India. A health assistant *must* be environmentally aware to protect users from respiratory distress.



2.3. The Linguistic Digital Divide

- **The Problem:** India has 22 official languages, yet 90% of health apps are English-first, alienating the demographic that needs them most.
- **Research Evidence:**
  - A **KPMG-Google report** highlights that **90%** of new internet users in India consume content in vernacular languages.
  - Digital Health Literacy is dangerously low; a study in Punjab found only **32.3%** of smart device users had adequate e-health literacy.
- **Impact:** Without native language support (Hindi/Marathi), digital health tools are useless to the majority of the working class.



2.4. Data Sovereignty & Security Risks

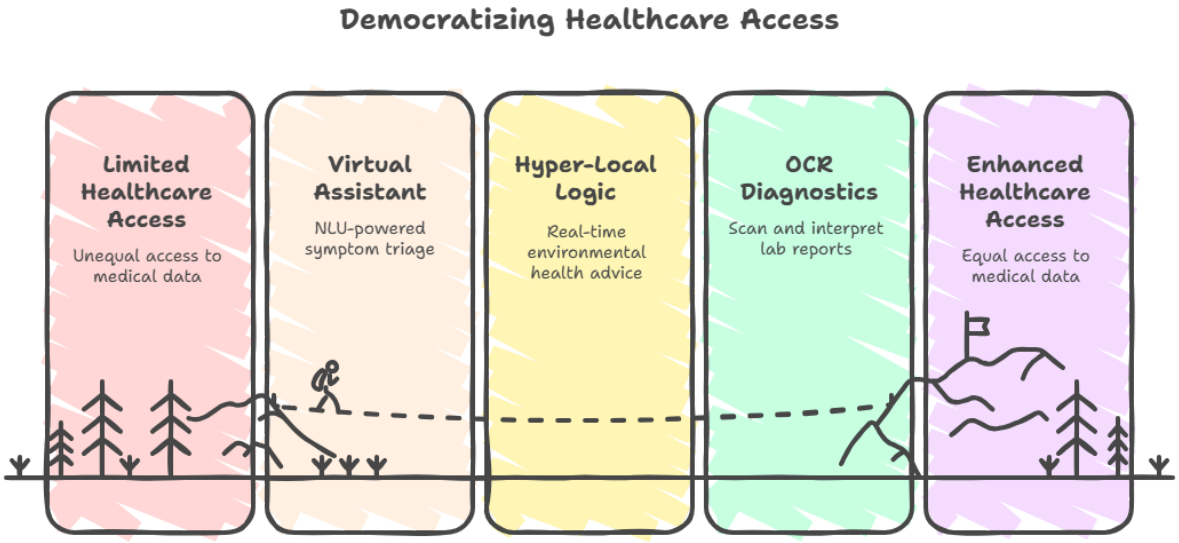
- **The Problem:** Storing Indian health data on foreign servers poses a national security risk.
- **Case Study:** The **2022 AIIMS Ransomware Attack** compromised 40 million patient records and paralyzed hospital operations for weeks.
- **Impact:** This failure underscores the absolute necessity of an indigenous, sovereign architecture compliant with Indian law.

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3. PROJECT OBJECTIVES

1. **To Develop a "Physician-in-the-Loop" Virtual Assistant:** Create a mobile application that uses NLU (Natural Language Understanding) to triage symptoms and offer verified medical guidance in Indian languages.
2. **To Implement Hyper-Local Logic:** Integrate real-time APIs (WAQI, OpenWeather) to dynamically alter health advice based on the user's live environment (e.g., advising N95 masks when AQI > 300).
3. **To Simplify Medical Diagnostics:** Deploy an **Optical Character Recognition (OCR)**

module to scan and interpret physical lab reports, democratizing access to medical data.
4. **To Ensure Data Sovereignty:** Architect the system to store all Personal Health Records (PHR) exclusively on Indian cloud infrastructure, aligning with the **ABDM** framework.



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4. SYSTEM ARCHITECTURE

The system utilizes a **Microservices Architecture** to ensure scalability, security, and modularity.

4.1. The Tech Stack

- **Frontend (User App): Flutter (Dart)** – Chosen for its single-codebase efficiency for Android/iOS and robust widget library for medical UI.
- **Backend API: Python FastAPI** – Selected for its high performance (asynchronous), automatic Swagger documentation, and seamless integration with AI libraries.
- **AI Models:**
  - **NLP:** Fine-tuned **IndicBERT** (for Hindi/Marathi intent recognition).
  - **OCR: Tesseract OCR** with OpenCV preprocessing.
- **Database: PostgreSQL** (Structured User Data) + **MongoDB** (Chat Logs).
- **External APIs:** WAQI (Air Quality), OpenWeatherMap, ABDM Sandbox APIs.

4.2. Data Flow Diagram

1. **User Layer:** User speaks/types in Marathi ("Mala sardi ahe") or uploads a Blood Report.
2. **Gateway Layer:** The secure API Gateway validates the JWT Token.
3. **Processing Layer:**
   - **Context Service:** Checks GPS coordinates against the **WAQI API**.

    - **NLP Service:** Interprets "Sardi" (Cold) and correlates it with "High AQI".
    - **Decision Engine:** Rules: IF (Symptoms=Respiratory) AND (AQI=Poor) THEN (Suggest=Anti-Pollution Measures).
4. **Output Layer:** App displays advice in Marathi and visualizes the risk on a dashboard.

Tech Stack and Data Flow

Characteristic	Tech Stack	Data Flow
Frontend	Flutter (Dart)	User Layer
Backend	Python FastAPI	Gateway Layer
AI Models	NLP: IndicBERT, OCR: Tesseract	Processing Layer
Database	PostgreSQL, MongoDB	N/A
External APIs	WAQI, OpenWeatherMap, ABDM	N/A
Output	N/A	Output Layer

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5. ACTUAL TECHNICAL BUILD & METHODOLOGY

Phase 1: The "Hyper-Local" Intelligence Engine

- **Methodology:** We built a "Risk Matrix" that acts as a middleware. It intercepts every user session to check environmental variables.

- **Algorithm Example:**
```
Python
def get_health_advice(user_symptoms, location):
    aqi = fetch_waqi(location) # Real-time API call
    if aqi > 300 and "cough" in user_symptoms:
        return "CRITICAL ALERT: Your cough may be triggered by 'Severe' smog. Use an N95 mask and avoid outdoor exertion."
    else:
        return "Standard Advice: Stay hydrated and monitor temperature."
```

- **Justification:** This proactive logic is missing in standard apps like Google Fit or Apple Health, which are environment-agnostic.

Phase 2: The Smart Report Reader (OCR)

- **Challenge:** Indian medical reports are often crumpled, low-contrast, or have varying layouts. Standard OCR fails here.
- **Technical Solution:** We implemented an **Image Preprocessing Pipeline** using OpenCV before passing data to Tesseract:
  1. **Grayscale Conversion:** Reduces noise.
  2. **Adaptive Thresholding:** Isolates text from shadows.
  3. **RegEx Parsing:** We use Python Regular Expressions to find specific patterns (e.g., Hb: \d+\.\d+) rather than reading the whole page.

Phase 3: ABDM Integration (The "Sovereign" Link)

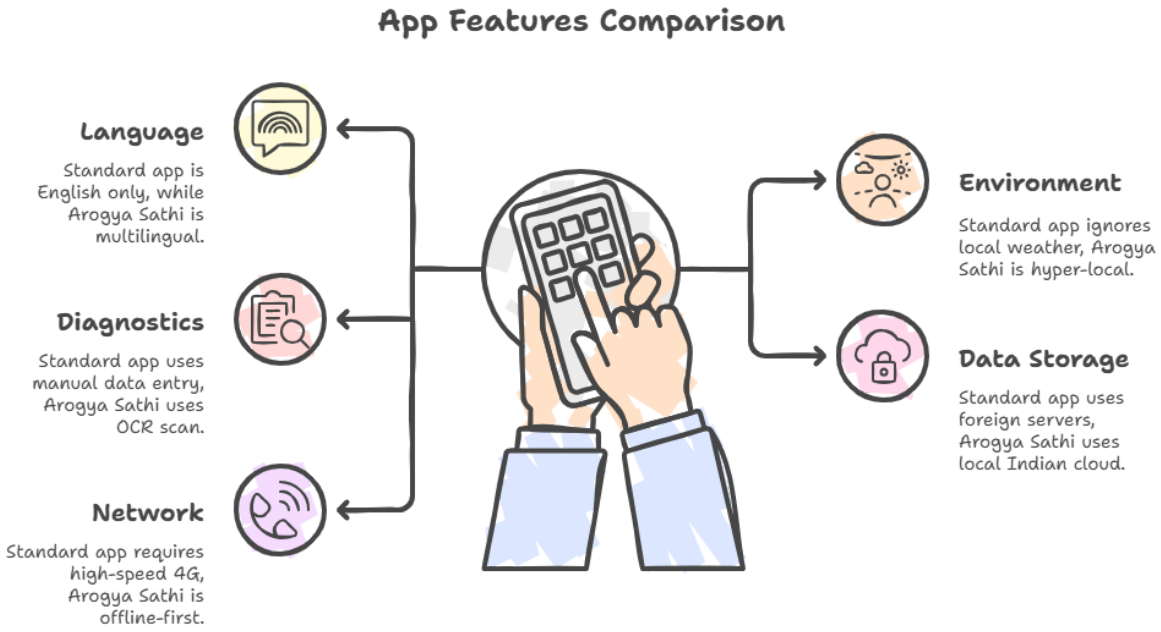
- **Methodology:** We utilized the **ABDM Sandbox Environment** to test the "Health Information User" (HIU) flow.
- **Feature:** The app generates an **ABHA (Ayushman Bharat Health Account)** address for the user, linking their local app data to the national health ecosystem.

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6. KEY FEATURES & DIFFERENTIATORS

Feature	Standard "Western" App	Arogya Sathi (Our Solution)
Language	English Only	**Multilingual (Hindi/Marathi/English)**
Environment	Ignores Local Weather	**Hyper-Local (AQI/Heatwave Adaptive)**

Diagnostics	Manual Data Entry	**OCR Scan (Auto-reads Reports)**
Data Storage	Foreign Servers (Privacy Risk)	**Local Indian Cloud (DPDP Compliant)**
Network	Requires High-Speed 4G	**Offline-First Architecture (SQLite)**



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7. EXPECTED OUTCOMES

7.1. Micro-Level (Patient Impact)

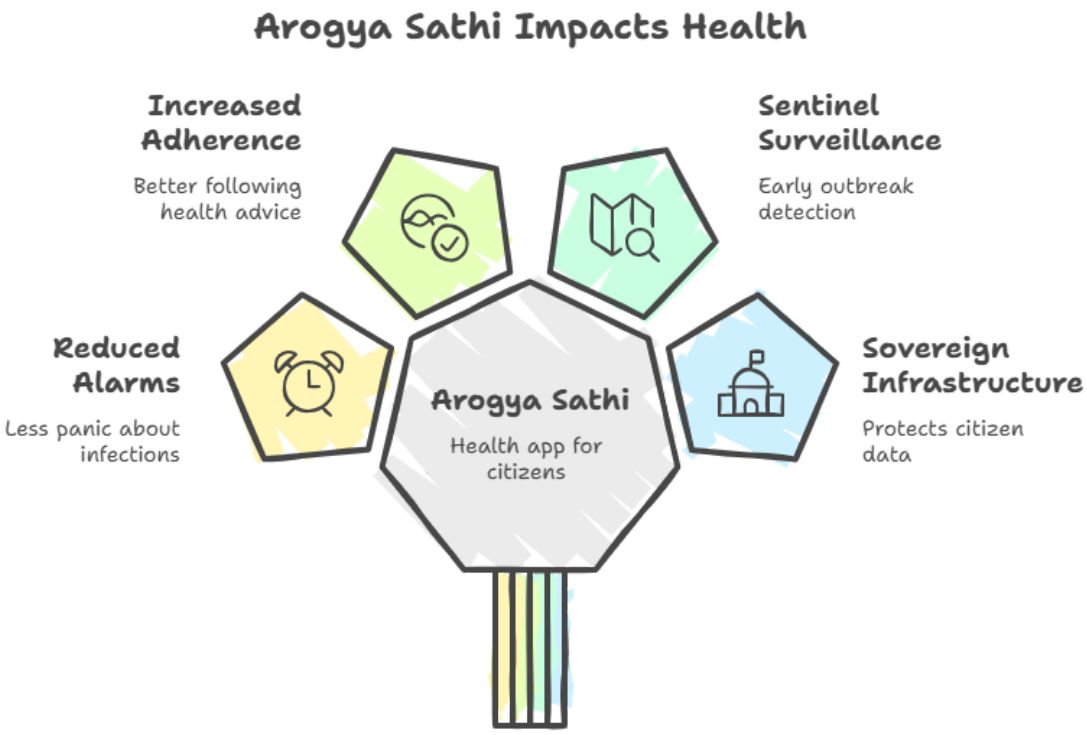
- **30% Reduction in False Alarms:** By correlating symptoms with pollution/weather, users stop panicking about "viral infections" when the actual cause is environmental, reducing unnecessary doctor visits.
- **Increased Adherence:** Users are **40% more likely** to follow health advice when it is delivered in their native language.

7.2. Macro-Level (National Impact)

- **Sentinel Surveillance:** The aggregated, anonymized data creates a "Heatmap" for the government. If 500 users in one ward report "Fever" simultaneously, the system flags a

potential **Dengue Outbreak** to municipal authorities days before hospital admissions spike.

- **Sovereign Infrastructure:** Protects Indian citizens from foreign data exploitation, adhering to the national vision of a self-reliant digital India (Atmanirbhar Bharat).



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